Pradhumn Pande

+91 9604151243 | pradyumpande@gmail.com | LinkedIn | Personal Website

Summary

AI/ML Engineer with **2+ years** of hands-on experience in building and deploying ML solutions using **Google Vertex AI** and scalable cloud data platforms. Skilled in ETL/ELT, data quality and validation, governance, CI/CD, and preparing trusted data for analytics, machine learning and AI-driven product features. Skilled in developing secure **ML pipelines**, implementing **AI-driven systems**, and enabling **enterprise AI adoption** through platform validation and data-driven insights.

Technical Skills

Languages:	Python, SQL, PySpark, NoSQL
Backend & Frameworks:	Node.js, Django, FastAPI, Unity, API gateways, Microservices Architecture.
Databases & Systems:	Data Pipelines, Distributed Database Systems, CI/CD pipelines, BigQuery.
Cloud & AI:	Google Cloud Platform (GCP), Vertex AI, Machine Learning models, LLM, Generative AI, Agentic AI, MLOps, Agent Evaluation metrics, Prompt Engineering, MCP Server Development ,Model Optimization ,Memory Management.
DevOps & Tools:	Git, GitHub, Docker, Postman, Azure DevOps, Microsoft Fabric

Experience

Accenture — Bangalore, India **Oct 2024 – Present**

Associate Data and AI Engineer

- Developed **Agentic AI** solutions that automated complex business processes, reducing execution time **from weeks to hours** through natural language interactions.
- Migrated CI/CD authentication from **HashiCorp Vault** to **Workload Identity Federation (WIF)**, enhancing security and eliminating static credentials.
- Developed **machine learning-powered monitoring** solutions, optimizing operations and cost by **up to 50%** and automating **30–40%** of workload effort for other teams.
- Built an **AI agent** and **Google Cloud Run** service that automated **80%** of secure access workflows for AD Group-managed data assets (**BigQuery** tables, datasets, buckets, and authorized views), eliminating manual access requests while ensuring **authorization and governance compliance**.
- Automated access requests and **TTL extension** for data stores such as **BigQuery** via an agent while maintaining workflow authorization and governance, eliminating **80–100 hours** of weekly manual effort across application teams.
- Collaborated with Microsoft teams to certify **Microsoft Fabric** for enterprise ML use cases through rigorous platform testing and validation, and recommended tooling for ML use cases.

Projects

Code Monitoring and Code Optimization | *Isolation Forest, Agentic AI, Azure AI Foundry, Power BI*

- Developed an **Isolation Forest**-based machine learning model to detect anomalies in storage account read transactions involving millions of requests. Integrated real-time anomaly reporting through **Power BI** dashboards and automated email notifications via **Microsoft Entra ID**, enabling proactive monitoring of abnormal cost spikes.
- Built an **AI-powered optimization agent** that analyzed detected anomalies and recommended efficient storage read strategies to reduce unnecessary transactions. The solution automated **60–70%** of optimization efforts, reduced storage read costs by **up to 50%**, and shortened the optimization cycle **from weeks to hours**.

Augmented Reality Based Navigation Interface System | *Unity, AR Core, AI Foundation and A* Algorithm*

- An **AR-based navigation system** offering immersive, real-time indoor guidance using **computer vision**, sensors, and metadata for accurate positioning.

Translate and Transcribe | *Node.js, API Development and JavaScript*

- An application that records and transcribes voice input and **translates it into multiple international languages** using third-party APIs.

Education

Savitribai Phule Pune University
B.Tech – Computer Science and Engineering
2020 – 2024 | CGPA: 8.8

XII (HSC)
Computer Science
2018 – 2020 | Percentage: 93.38

Certifications & Publication

- Generative AI Leader Certification
- Associate Cloud Engineer Certificate
- MERN Stack Bootcamp 2023 – MongoDB, Express, React and NodeJS
- Certified Partner Specialist Gemini Enterprise Development and Deployment
- Claude Certified Developer - Anthropic
- Publication - Augmented Reality Based Navigation Interface System (2024)